

ArchiShield is an AI-powered tool that detects, measures, and helps fix data loss in IFC (BIM) models across 5 integrity levels, with a role-adaptive interface and a natural-language AI assistant. This document covers architecture, tech stack, modules, data flow, and setup.

01 Architecture & Tech Stack

Frontend	Streamlit (Python) — multipage app, widgets, session state
IFC Engine	ifcopenshell — parses and traverses IFC entities/relationships
AI / NLQ	Groq API (Llama 3.3 70B) — natural-language Q&A; grounded in scanned model data
Storage	Supabase Storage — private per-user IFC file storage
Reports	fpdf + pandas — PDF/CSV report generation
Voice Input	Browser Web Speech API — client-side, no extra backend

Browser



Streamlit



ifcopenshell



Analysis + Groq + Supabase

02 Module Breakdown

Home Dashboard	Login, IFC upload, runs the 5-level scan, role-adapted summary view
Proxy Classification	Flags IfcBuildingElementProxy entities as Valid / Invalid / Unknown
Pset Analysis	Checks required Property Set completeness per element type
3D Viewer / Heatmap	3D model view + 2D spatial issue density overlay
Storey Quality	Per-floor breakdown of element counts and issue ratios
NBC Compliance	Section-by-section check against National Building Code 2016
Corrections	Lists each issue with a suggested fix
Model Score	Weighted 0-100 quality score with severity label
BCF / Version Compare	Exports issues as .bcf; diffs two IFC versions
Cloud Library	Upload / list / download / delete IFC files (Supabase)
Ask Your Model	Voice + text NLQ assistant grounded in live model data

03 5-Level Data Loss Pipeline

L1 Semantic	Elements exported as generic IfcBuildingElementProxy (no real type)
L2 Property	Missing required Pset (e.g. Pset_WallCommon) for the element type
L3 Quantity	Missing area / volume / length quantity data

L4 Relationship	Element not linked to a building storey
L5 Geometry	Element has no shape/geometry representation at all

$quality_score = 100 - \sum (level\ loss\ \% \times level\ weight)$ · Severity: LOW ≥85, MEDIUM ≥70, HIGH ≥50, else CRITICAL

04 AI Assistant — Grounding & Token Management

- Live IFC file is re-scanned per session (cached) — not just the cached summary dict.
- Compact JSON sent to Groq: totals, per-storey breakdown, element type counts, up to 40 issue-flagged elements (proxies / missing Psets / missing fire rating prioritised first).
- System prompt restricts answers to ONLY the supplied data — explicitly told never to invent numbers.
- Context capped (40 elements, 20 storeys, 25 element types, compact JSON) to stay under Groq's free-tier 12,000 tokens/min limit — typical request is ~1,800 tokens.
- Voice input via browser Web Speech API — Chrome/Edge, no extra package.

05 Role-Adaptive Interface

Architect	Violet — minimal, visual-first, large score display
BIM Manager	Blue — dense technical dashboard, full IFC detail
Contractor	Orange — bold action board, fix-first call-to-actions
Facility Manager	Green — checklist / compliance-tracker style
Student/Researcher	Cyan — learning mode with plain-English explanations

All pages also support a **Technical / Business / Full** view-mode toggle, independent of role, switching between IFC jargon and plain-English descriptions.

06 Setup & Installation

```

pip install streamlit ifcopenshell pandas fpdf groq supabase

# Get a free key: console.groq.com/keys
setx GROQ_API_KEY "your-key-here" (windows – reopen terminal after)
export GROQ_API_KEY="your-key-here" (Mac/Linux)

streamlit run Home.py

```

07 Known Limitations

- NLQ element-level detail capped to 40 issue-flagged elements — not exhaustive for very large models.
- Proxy classification is rule-based (name-pattern matching); ML classifier (LightGBM) is roadmapped, not yet built.
- Team/collaborative workspaces are designed but not wired into this prototype.
- Role selection is session-based for this demo; full Google OAuth is planned for the ArchiTechs platform.